

Navigating the Pareto Frontier of Alignment: Spectrum-Adaptive Fine-Tuning for LLMs

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Abstract

Supervised Fine-Tuning with Negative Log-Likelihood (NLL) remains the standard post-training paradigm for Large Language Models, yet it imposes a disproportionately aggressive update force on low-probability target tokens. This focus forces the model to prioritize minimizing the loss of difficult samples over optimizing the overall quality of the generation, often leading to unwarranted overconfidence. On the other hand, alternatives like Dynamic Fine-Tuning suffer from vanishing gradients on these tokens, which severely hinders the acquisition of new concepts. To bridge this gap, we propose **Spectrum-Adaptive Fine-Tuning (SAFT)**, a unified framework that interpolates between the aggressive learning signal of NLL and the robust nature of probability-weighted optimization. By adaptively balancing these objectives, SAFT effectively mitigates outlier sensitivity without sacrificing learning efficiency. Empirically, our method achieves state-of-the-art performance on mathematical reasoning benchmarks, demonstrating superior generalization on out-of-distribution tasks. Furthermore, evaluations on general conversational alignment validate SAFT’s broad adaptability across diverse data regimes. Our code is available at <https://github.com/sjtu-scx/SAFT>.

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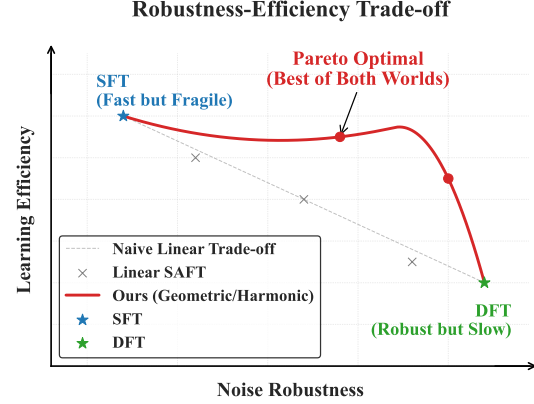


Figure 1. The Empirical Pareto Frontier. We visualize the trade-off between Learning Efficiency (Y-axis, **upwards** indicates higher efficiency) and Noise Robustness (X-axis, **rightwards** indicates stronger resistance). SFT and DFT occupy opposing ends: “Fast but Fragile” vs. “Robust but Slow.” Crucially, simple interpolation via Lin-SAFT yields a suboptimal compromise falling below the frontier. In contrast, our non-linear strategies traverse the Pareto frontier, pushing the performance boundary outward to achieve the “Best of Both Worlds” outcome.

1. Introduction

Post-training is a critical stage that transforms broad pre-trained representations into reliable, task-oriented intelligence. Among post-training paradigms, Supervised Fine-Tuning (SFT) remains the dominant approach for aligning Large Language Models (LLMs) (Wei et al., 2022; Zhou et al., 2023; Ouyang et al., 2022; Touvron et al., 2023), achieving strong performance across domains such as mathematics (Li et al., 2024; Ye et al., 2025) and code generation (Luo et al., 2024). Standard SFT operates by maximizing the likelihood of ground-truth tokens, implicitly assigning a gradient weight of $1/p_t$ to each token, where p_t is the model’s predicted probability. While effective, this objective is inherently hypersensitive to low-confidence predictions. As $p_t \rightarrow 0$, this relative weight scales aggressively, forcing the model to disproportionately focus on these hard tokens. Consequently, the model tends to overfit to noisy data or outliers at the expense of generalizable patterns (O’Mahony et al., 2024; Kim et al., 2025; Lin et al.,

2024).

To mitigate this instability, recent works such as Dynamic Fine-Tuning (DFT) (Wu et al., 2025), propose removing the denominator entirely, yielding a gradient proportional to ∇p_t . By effectively capping the gradient weight, DFT acts as a robust learner that ignores low-confidence noise. However, we argue that SFT and DFT represent two extreme discrete points in the optimization landscape: SFT is overly aggressive on uncertainty, while DFT may be overly conservative, potentially discarding valid learning signals embedded in difficult reasoning steps.

To systematically explore the space between these extremes, we visualize the optimization landscape in Figure 1. This plot reveals an empirical **Pareto frontier** (Miettinen, 1999; Messac et al., 2003) governing the trade-off between *Noise Robustness* and *Learning Efficiency* (Lin et al., 2019; Abbass et al., 2001). Standard SFT and DFT occupy the opposing ends of this spectrum: SFT maximizes efficiency but collapses under noise, while DFT ensures robustness at the cost of slow convergence. Our analysis exposes a critical gap: simple interpolation yields a suboptimal compromise that falls below the frontier. Although such a linear scheme numerically connects the SFT and DFT endpoints, it only attenuates the low-confidence over-updating inherited from maximum likelihood, without eliminating the non-vanishing influence of extremely low-probability tokens. In contrast, non-linear strategies are required to reshape the low-confidence geometry and traverse the frontier more effectively. Ideally, we seek methods that can adaptively balance the “stiffness” of the gradient against the noise level of the data.

In this paper, we propose **Spectrum-Adaptive Fine-Tuning (SAFT)**, a unified framework that bridges the gap between log-likelihood maximization (SFT) and probability-weighted optimization (DFT). The term *Spectrum-Adaptive* captures the dual nature of our approach: First, rather than treating SFT and DFT as binary, mutually exclusive choices, SAFT establishes a **continuous optimization spectrum** parameterized by α . This continuity allows us to precisely navigate the trade-off between the aggressive learning efficiency of SFT and the robust noise resilience of DFT. Second, we interpret the token probability p_t as a confidence spectrum. SAFT *adaptively* modulates the gradient weight according to where a token falls within this range. This mechanism acts as a soft filter that selectively modulates gradients: dampening those from low-confidence predictions (often pure noise) while preserving strong signals from high-confidence, valid reasoning steps.

We instantiate this framework with three interpolation strategies: **Linear**, **Geometric**, and **Harmonic**. While Linear Interpolation (Lin-SAFT) serves as a canonical baseline, we demonstrate that non-linear strategies are crucial for optimal

performance. Through rigorous analysis, we show that these strategies offer distinct gradient dynamics: Geometric Interpolation (Geo-SAFT) maintains high learning efficiency, making it superior for clean, high-quality data. Conversely, Harmonic Interpolation (Har-SAFT) demonstrates stronger robustness, effectively balancing the rejection of noise with the retention of difficult-but-valid training signals. This makes Har-SAFT particularly effective in noisy regimes where discerning “hard” from “noisy” is critical.

Our contributions are threefold:

- We provide a rigorous theoretical explanation for the trade-off between SFT and DFT, grounding their gradient behaviors within a unified weighting framework.
- To bridge this gap, we propose Spectrum-Adaptive Fine-Tuning (SAFT), a framework that introduces a continuous optimization spectrum between SFT and DFT. Specifically, SAFT encompasses three principled interpolations: Linear, Geometric, and Harmonic.
- Extensive experiments across diverse LLM architectures (Qwen, DeepSeek) and scales (1.5B/7B/14B) demonstrate that SAFT achieves state-of-the-art performance and significantly enhances out-of-domain (OOD) generalization.

2. Related Work

LLM Post-training. Post-training is key for aligning LLMs with downstream tasks, mainly through Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL). SFT is widely used for its simplicity and effectiveness, serving as imitation learning (Hussein et al., 2017; Osa et al., 2018) to align models with expert distributions (Zhou et al., 2023; Chen et al., 2024). However, it treats all training samples equally, lacking the flexibility to discriminate against noise, which can lead to overfitting erroneous patterns in imperfect datasets (Chu et al., 2025; Li et al., 2025b). RL uses algorithms like PPO (Schulman et al., 2017) and GRPO (Shao et al., 2024) to maximize cumulative rewards via online exploration but faces issues with training instability and high computational costs (Engstrom et al., 2020; Zheng et al., 2023).

Optimization Objectives in SFT. SFT typically minimizes Negative Log-Likelihood (NLL), equivalent to minimizing the KL divergence (Cover, 1999) between the generated distribution and the target distribution. Recent work (Li et al., 2025a) argues that NLL induces rigid token-wise fitting, which can overfit intermediate Chain-of-Thought (CoT) noise and distract the model from final-answer correctness. DFT (Wu et al., 2025) addresses this issue by optimizing a smooth approximation of inference accuracy (i.e., $-p$), using the predicted probability p to counteract NLL’s

gradient bias. Building on DFT, Anchored Supervised Fine-Tuning (ASFT) (Zhu et al., 2026a) introduces KL-based anchoring to mitigate distributional drift while retaining its underlying probability-weighted optimization. However, DFT-style probability weighting still strongly suppresses low-probability tokens, which can hinder the acquisition of difficult but informative knowledge. In contrast, SAFT directly reshapes the probability-dependent weighting profile to balance noise robustness with efficient learning from low-probability signals.

Token Reweighting in SFT. Standard SFT weights all tokens uniformly despite differences in difficulty and information content. Recent methods broadly fall into two categories. Noise Suppression methods (Wu et al., 2025; Su et al., 2025; Zhu et al., 2026b) down-weight low-probability tokens through masking or thresholding, but may also discard valid hard examples and hinder complex reasoning (Kim et al., 2025). Heuristic Adaptive methods (Ruan et al., 2025; Li et al., 2025a; Leng et al., 2022) use auxiliary signals such as gradient variance or causal influence, often adding computational cost and hyperparameter sensitivity. In contrast, SAFT bridges SFT and DFT through a unified gradient framework. Its spectrum-adaptive interpolation suppresses unreliable low-confidence signals while preserving useful hard examples, balancing robustness and plasticity without auxiliary heuristics.

3. Preliminaries

In this section, we unify SFT and DFT under a gradient-based framework and empirically demonstrate a critical trade-off between noise robustness and learning efficiency, motivating our proposed approach.

3.1. A Unified Gradient Perspective

Consider a supervised dataset $\mathcal{D} = \{(x, y^*)\}$ consisting of input prompts x and their corresponding ground-truth reference responses y^* . Let $p_t = \pi_\theta(y_t^* | y_{<t}^*, x)$ be the conditional probability assigned by the policy model to the ground-truth correct token at generation step t .

Supervised Fine-Tuning (SFT). The standard SFT objective minimizes the negative log-likelihood over the training dataset: $\mathcal{L}_{\text{SFT}} = -\sum_t \log p_t$. Analyzing its gradient reveals an implicit token-level weighting mechanism:

$$\nabla \mathcal{L}_{\text{SFT}} = -\sum_{t=1}^T \underbrace{\frac{1}{p_t}}_{\text{Coefficient } \gamma_t} \nabla_\theta p_t. \quad (1)$$

We term $\gamma_t = 1/p_t$ the **Effective Gradient Coefficient**. It acts as an aggressiveness factor: as the model’s confidence drops ($p_t \rightarrow 0$), this relative multiplier scales aggressively ($\gamma_t \rightarrow \infty$). This hypersensitivity accelerates learning

Table 1. Robustness Analysis. Performance of Qwen2.5-Math-1.5B at 50% noise ($\eta = 0.5$). SFT collapses below the base model, while DFT remains resilient.

Method	Noise η	Avg. Acc (%)	Δ vs. Base
Base Model	-	11.00	-
SFT	0.5	5.32	-5.68
DFT	0.5	24.01	+13.01

on “hard” samples (where p_t is initially low), but renders the model vulnerable to outliers and label noise, causing gradient instability when the target is erroneous.

Dynamic Fine-Tuning (DFT). To mitigate this, Wu et al. (2025) proposed maximizing the raw likelihood expectation, yielding $\nabla \mathcal{L}_{\text{DFT}} \propto -\sum 1 \cdot \nabla_\theta p_t$. From the perspective of Eq. 1, DFT effectively clamps the gradient coefficient to a constant $\gamma_t = 1$. While this bounded coefficient prevents excessive over-penalization on noisy tokens, it treats all tokens equally regardless of difficulty. Consequently, it dampens the learning signal for complex, high-value reasoning steps where the model naturally struggles (low p_t), leading to inefficiency.

3.2. The Vulnerability of SFT Under Noise

We first investigate robustness by evaluating LLMs under severe supervision noise. We employ a Target Permutation protocol and introduce Noisy Dataset. Specifically, for a noise level $\eta = 0.5$, we randomly shuffle the ground-truth solutions among 50% of the training samples, pairing inputs with logically irrelevant answers.

As shown in Table 1, SFT suffers catastrophic degradation, with its performance (5.32%) dropping significantly below even the base model (11.0%). This negative gain confirms that the unbounded $1/p_t$ weight forces the model to memorize erroneous mappings rather than ignoring them. In contrast, DFT maintains substantial reasoning capabilities (24.01%), demonstrating that its constant weight mechanism naturally filters out the gradient impact of low-probability, mismatched targets.

3.3. The Efficiency Bottleneck of Conservative Optimization

While DFT offers robustness by bounding the effective gradient coefficient ($\gamma = 1$), this conservatism comes at a cost: inefficiency on hard, high-value data.

Consider a high-quality reasoning dataset (e.g., LIMO (Ye et al., 2025)) where data points represent complex logical steps. Ideally, a model should rapidly adapt to these “hard positives” (where the initial probability p_t is low). Standard SFT intrinsically accelerates this process via its unbounded coefficient ($\gamma \propto 1/p_t$), which amplifies the learning signal

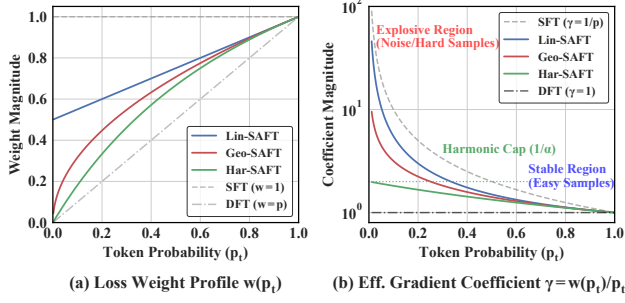


Figure 2. Gradient Weight Scaling and Effective Gradient Coefficient Profiles. We compare different SAFT interpolation strategies under $\alpha = 0.5$. (a) The left plot shows the loss weight $w(p_t)$ as a function of the target-token probability p_t . Linear interpolation preserves a non-zero floor as $p_t \rightarrow 0$, while Geo-SAFt and Har-SAFt both decay to zero in the low-probability region. (b) The right plot shows the effective gradient coefficient $\gamma_t = w(p_t)/p_t$. SFT exhibits an unbounded coefficient as $p_t \rightarrow 0$, DFT remains constant, and the SAFT variants interpolate between these regimes: Geo-SAFt is unbounded in the low-probability limit, whereas Har-SAFt is upper-bounded by $1/\alpha$.

for unknown patterns. Although this often leads to instability (as discussed in Sec. 3.2), the *mechanism* itself provides the necessary “force” to fit complex distributions quickly.

The Cold Start Problem. In contrast, DFT treats all tokens equally. For a difficult reasoning step where $p_t \approx \epsilon$, the gradient magnitude in DFT scales linearly with the probability ($\|g\| \propto p_t \approx \epsilon$). As we formally derive in Appendix C.2, this leads to a Hyperbolic Stagnation phase: the time required to learn a hard pattern scales inversely with its difficulty ($T \propto 1/\epsilon$). On pristine, information-rich datasets, this dampening effect results in significant under-utilization of data. The model “sees” and processes the complex reasoning steps in the training data but lacks the sufficient gradient magnitude to effectively prioritize learning them, leading to slow convergence or failure to capture crucial long-tail knowledge patterns.

The Dilemma. We identify a fundamental conflict: SFT offers efficiency but lacks robustness, while DFT offers robustness but suffers from slow convergence. This motivates our proposed **SAFT**, a unified framework that adaptively interpolates between these regimes to achieve both noise resilience and efficient learning.

4. Methodology

4.1. SAFT: Spectrum-Adaptive Fine-Tuning

Recall from Section 3 that the core conflict lies in the effective gradient coefficient γ_t applied to the likelihood gradient $\nabla_{\theta} p_t$. SFT imposes an aggressive, unbounded coefficient ($\gamma_{\text{SFT}} = 1/p_t$), while DFT enforces a conservative, constant coefficient ($\gamma_{\text{DFT}} = 1$).

Algorithm 1 SAFT Training Procedure

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1: Input: Dataset  $\mathcal{D}$ , Model  $\pi_{\theta}$ , Interpolation type  $\mathcal{T} \in \{\text{Lin}, \text{Geo}, \text{Har}\}$ , Parameter  $\alpha$ , Learning rate  $\eta$ .
2: for each batch  $(x, y^*) \sim \mathcal{D}$  do
3:   Compute logits and probabilities  $p_t \leftarrow \pi_{\theta}(y_t^* | y_{<t}^*, x)$  for all  $t$ .
4:   Calculate weight  $w_t$  based on  $\mathcal{T}$ :
5:   if  $\mathcal{T} = \text{Linear}$  then
6:      $w_t \leftarrow \alpha \cdot p_t + (1 - \alpha)$ 
7:   else if  $\mathcal{T} = \text{Geometric}$  then
8:      $w_t \leftarrow p_t^{\alpha}$ 
9:   else if  $\mathcal{T} = \text{Harmonic}$  then
10:     $w_t \leftarrow p_t / (\alpha + (1 - \alpha)p_t)$ 
11:   end if
12:   Detach weight from graph:  $w_t \leftarrow \text{detach}(w_t)$ .
13:   Compute Loss:  $\mathcal{L} \leftarrow -\frac{1}{T} \sum_t w_t \log p_t$ .
14:   Update  $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ .
15: end for

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We propose **SAFT**, a framework that smoothly interpolates between these two regimes via a control parameter $\alpha \in [0, 1]$. We achieve this by introducing a probability-dependent weight $w(p_t, \alpha)$ into the loss function:

$$\mathcal{L}_{\text{SAFT}}(\theta; \alpha) = \mathbb{E} \left[- \sum_{t=1}^T \text{sg} \left(w(p_t, \alpha) \right) \log p_t \right]. \quad (2)$$

Here, $\text{sg}(\cdot)$ is the stop-gradient operator, which ensures that w acts solely as a gradient modulator without introducing additional complex derivative terms during backpropagation. Crucially, this design leads to an effective gradient coefficient that becomes:

$$\gamma_{\text{SAFT}} = \frac{w(p_t, \alpha)}{p_t}. \quad (3)$$

While γ_{SAFT} characterizes the relative amplification with respect to $\nabla_{\theta} p_t$, it is not the actual gradient magnitude on the model logits. Since the weight $w(p_t, \alpha)$ is detached, the weighted NLL loss yields the following gradient magnitude with respect to the target token logit z_t :

$$\left| \frac{\partial \mathcal{L}_{\text{SAFT}}}{\partial z_t} \right| = w(p_t, \alpha)(1 - p_t). \quad (4)$$

We refer to this quantity as the *target-logit influence*, since it measures how strongly a training token pushes the model to increase the logit of the target token. This distinction is important: a singular effective coefficient does not necessarily imply an unbounded logit-space gradient, because the softmax Jacobian cancels part of the $1/p_t$ singularity. Therefore, beyond interpolating between γ_{SFT} and γ_{DFT} , a desirable interpolation should also make the target-logit influence of extremely low-confidence tokens vanish, i.e.,

$$w(p_t, \alpha)(1 - p_t) \xrightarrow{p_t \rightarrow 0} 0. \quad (5)$$

Our goal is to design $w(\cdot)$ such that SAFT can navigate the SFT–DFT spectrum while avoiding the undesirable low-confidence behavior of naive linear interpolation. We explore three distinct interpolation profiles based on their asymptotic behavior. The complete, easy-to-implement training pipeline integrating these strategies is summarized in Algorithm 1.

1. Linear Interpolation (Lin-SAFT). A natural yet limited choice is to linearly mix the SFT and DFT weights:

$$w_{\text{lin}}(p_t, \alpha) = (1 - \alpha) + \alpha p_t. \quad (6)$$

Substituting this into Eq. (3), we obtain

$$\gamma_{\text{lin}}(p_t, \alpha) = \frac{(1 - \alpha) + \alpha p_t}{p_t} = \frac{1 - \alpha}{p_t} + \alpha. \quad (7)$$

Thus, for any $\alpha < 1$, the effective coefficient remains singular as $p_t \rightarrow 0$. More importantly, its target-logit influence satisfies

$$w_{\text{lin}}(p_t, \alpha)(1 - p_t) = [(1 - \alpha) + \alpha p_t](1 - p_t) \xrightarrow{p_t \rightarrow 0} 1 - \alpha. \quad (8)$$

Therefore, simple linear interpolation only attenuates the CE-like low-confidence update, but still leaves a non-vanishing influence for extremely low-probability tokens. It numerically connects SFT and DFT, yet fails to fundamentally remove the interference of such tokens during training.

2. Geometric Interpolation (Geo-SAFT). To overcome this limitation, we introduce a nonlinear interpolation with power-law geometry:

$$w_{\text{geo}}(p_t, \alpha) = p_t^\alpha. \quad (9)$$

Substituting this into Eq. (3), the effective coefficient becomes

$$\gamma_{\text{geo}}(p_t, \alpha) = \frac{p_t^\alpha}{p_t} = \frac{1}{p_t^{1-\alpha}}. \quad (10)$$

For $0 < \alpha < 1$, γ_{geo} retains a “soft” singularity. Thus, Geo-SAFT is geometrically closer to SFT: it preserves stronger learning signals in the low-to-middle confidence region and is therefore more efficiency-oriented. At the same time, unlike Lin-SAFT, Geo-SAFT is conservative at the extreme low-confidence limit:

$$w_{\text{geo}}(p_t, \alpha)(1 - p_t) = p_t^\alpha(1 - p_t) \xrightarrow{p_t \rightarrow 0} 0. \quad (11)$$

Thus, Geo-SAFT suppresses the influence of extremely low-probability tokens while still retaining more SFT-like plasticity before reaching the extreme tail.

3. Harmonic Interpolation (Har-SAFT). To obtain a more safety-oriented geometry, we consider harmonic interpolation:

$$w_{\text{har}}(p_t, \alpha) = \frac{p_t}{\alpha + (1 - \alpha)p_t}. \quad (12)$$

Table 2. Robustness to Label Noise on Qwen2.5-Math-1.5B. Performance comparison at 50% label noise ($\eta = 0.5$) injected via target permutation. Δ denotes the difference from the clean baseline. Best results are in bold, second best are underlined.

Method	Avg. Acc (%)	Δ vs. Base
Base (Clean, $\eta = 0$)	11.0	–
SFT	5.32	-5.68
Lin-SAFT ($\alpha = 0.5$)	8.35	-2.65
Geo-SAFT ($\alpha = 0.5$)	20.07	+9.07
Har-SAFT ($\alpha = 0.5$)	23.41	+12.41
DFT	<u>24.01</u>	<u>+13.01</u>
Geo-SAFT ($\alpha = 2.0$)	24.11	+13.11

Analysis of Eq. (3) gives

$$\gamma_{\text{har}}(p_t, \alpha) = \frac{1}{\alpha + (1 - \alpha)p_t} \leq \frac{1}{\alpha}. \quad (13)$$

Therefore, Har-SAFT is geometrically closer to DFT: its effective coefficient is explicitly bounded, providing a safety cap for low-confidence updates. Moreover, its target-logit influence also vanishes in the extreme low-confidence limit:

$$w_{\text{har}}(p_t, \alpha)(1 - p_t) = \frac{p_t(1 - p_t)}{\alpha + (1 - \alpha)p_t} \xrightarrow{p_t \rightarrow 0} 0. \quad (14)$$

Compared with Geo-SAFT, Har-SAFT is more conservative in the low-confidence region and is therefore better suited when such tokens are more likely to be noisy, unreliable, or too far from the current model distribution.

4.2. Analysis of Gradient Dynamics and Regime Selection

The above asymptotic behaviors show that Geo-SAFT and Har-SAFT both address the interference of extremely low-probability tokens, since their logit-space influence vanishes as $p_t \rightarrow 0$. However, they encode different geometric biases along the SFT–DFT spectrum. Geo-SAFT has an effective-gradient curve closer to SFT and therefore favors learning efficiency, while Har-SAFT has an effective-gradient curve closer to DFT and provides an explicit safety cap, thereby favoring safety and robustness.

The Efficiency Regime (Clean / Hard Data). In datasets with high SNR, low-probability tokens are more likely to represent learnable hard examples rather than noise. Standard DFT ($w = p_t$) may suppress these samples too aggressively, leading to under-fitting. Geo-SAFT addresses this through its SFT-like soft singularity: although its influence vanishes at the extreme tail, it retains stronger learning signals in the low-to-middle confidence region. As seen in Figure 2b (red curve), Geo-SAFT remains closer to SFT than Har-SAFT in its effective-gradient profile, which helps accelerate convergence and capture long-tail knowledge that

DFT would under-utilize. Thus, Geo-SAFT is preferred when the data is signal-dominant. We empirically validate Geo-SAFT’s superiority in this high-SNR conversational regime in Appendix H.

The Safety Regime (Noisy Data / Reasoning). As illustrated in Figure 2b (green curve), Har-SAFT maintains a bounded effective gradient coefficient, saturating at $1/\alpha$ as $p_t \rightarrow 0$, in sharp contrast to the soft singularity of Geo-SAFT. This bounded profile yields a linearly vanishing target-logit influence ($O(p_t)$), providing stronger suppression of low-confidence (high-uncertainty) tokens. This design ensures that, compared to SFT and Geo-SAFT, the gradient influence of potentially erroneous or noisy data points vanishes more rapidly as the model’s confidence declines, directly contributing to stronger noise robustness.

This theoretical advantage is further corroborated in Table 2, where Har-SAFT significantly outperforms SFT and Lin/Geo-SAFT variants under heavy label noise ($\eta = 0.5$). Specifically, Har-SAFT achieves an average accuracy of 23.41%, far surpassing SFT (5.32%), Lin-SAFT (8.35%), and Geo-SAFT (20.07%), and closely rivals DFT and its high-order variants. More comprehensive results across different noise levels are provided in Table 6 in the Appendix. These results empirically validate that Har-SAFT’s “Safe Alignment” mechanism effectively filters out noisy and misleading supervision, making it the preferred choice in low-SNR or highly sensitive reasoning tasks.

Practical Pre-Test Selection Protocol. In real-world scenarios, the intrinsic Signal-to-Noise Ratio (SNR) of a dataset is often unknown *a priori*. To strictly prevent test-set overfitting and avoid computationally expensive trial-and-error, we establish a lightweight pre-test selection criterion using a held-out validation split. Specifically, practitioners can train SFT and DFT baselines on a small subset of the training data (e.g., 5%) and evaluate them on a standard validation set.

- **Efficiency Regime (Geo-SAFT):** If SFT outperforms DFT on the validation set, the data is empirically treated as *Signal-Dominant*. Geo-SAFT should be selected because its sub-linear geometry remains closer to SFT, preserving the learning efficiency required for hard but clean examples.
- **Safety Regime (Har-SAFT):** If DFT outperforms SFT, the data is treated as *Noise-Dominant*. Har-SAFT should be selected because its strictly linear decay ($O(p)$) enforces an explicit safety cap (the “Safe Corridor”), preventing catastrophic overfitting to noise.

As demonstrated in Appendix D, this validation-based preference is highly consistent with downstream test performance, providing a computationally inexpensive and rigorous standard for method selection before any test evaluation.

Through this data-adaptive geometric selection, SAFT can dynamically match different noise regimes, achieving a superior robustness–efficiency trade-off.

5. Experiments

5.1. Experimental Setup

Models and Datasets. We validate our method across diverse model families at various scales: Qwen series (Qwen2.5-Math-1.5B, Qwen2.5-Math-7B (Team et al., 2024), Qwen3-8B, Qwen3-14B (Yang et al., 2025)) and DeepSeekMath-7B (Shao et al., 2024). All models are trained on 100K randomly sampled instances from NuminaMath-CoT (Li et al., 2024).

Implementation Details. We utilize the verl framework (Sheng et al., 2025) for all experiments. All models are trained for one epoch on the sampled dataset. For SAFT variants, we set the interpolation parameter $\alpha = 0.5$ by default, where $\alpha \in [0, 1]$ controls the trade-off between SFT ($\alpha = 0$) and DFT ($\alpha = 1$). Optimization is performed using AdamW (Loshchilov & Hutter, 2019) with a global batch size of 256, achieved through a micro-batch size of 4 per GPU with gradient accumulation, and a maximum input length of 2048 tokens. The initial learning rate is set to 5×10^{-5} for all models, following a cosine decay schedule with a warm-up ratio of 0.1.

Evaluation Protocol. We evaluate on six mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021), Minerva Math (Lewkowycz et al., 2022), Olympiad-Bench (He et al., 2024), AIME 2024 (American Institute of Mathematics, 2024), AIME 2025 (American Institute of Mathematics, 2025), and AMC 2023 (Mathematical Association of America, 2023), using the Qwen2.5-Math pipeline (Qwen Team et al., 2024). All models use chain-of-thought (CoT) prompting for step-by-step reasoning. Following Luffy (Yan et al., 2026), we report avg@32 for AIME/AMC (small test sets) and pass@1 for others, with temperature 1.0 and max length 4096. For out-of-domain evaluation, we test on ARC-c (Clark et al., 2018) (open-domain reasoning), GPQA-diamond (Rein et al., 2024) (graduate-level science, GPQA*), and MMLU-Pro (Wang et al., 2024) (academic reasoning), with shuffled options to prevent contamination.

Noise Injection Protocol. To evaluate robustness, we construct the Noisy-Numina benchmarks by applying a Target Permutation protocol to the clean subset. For a given noise ratio $\eta \in \{0.1, 0.2, 0.3, 0.5\}$, we randomly shuffle the ground-truth solutions among $\eta \times 100\%$ of the training samples. Crucially, to prevent models from identifying noise via format artifacts, we atomically swap all target-related

Table 3. **In-Domain Mathematical Reasoning Performance.** We compare SFT, DFT, and SAFT variants (Linear, Geometric, Harmonic) across Qwen2.5-Math (1.5B/7B), Qwen3 (8B/14B), and DeepSeek-Math-7B backbones. **Bold** indicates the best result per backbone, underline indicates the second best.

Backbone	Method	MATH	Minerva	Olympiad	AIME'24	AIME'25	AMC	Avg.
DeepSeek-Math-7B	Base	4.40	0.74	0.89	0.00	1.00	1.17	1.22
	SFT	30.40	11.76	8.00	0.21	0.21	9.94	10.09
	DFT	<u>43.20</u>	16.91	13.19	1.56	1.67	18.79	<u>15.89</u>
	Lin-SAFT	36.20	13.24	9.33	0.42	0.00	12.12	11.88
	Geo-SAFT	41.60	19.49	<u>14.22</u>	1.56	0.52	15.02	15.40
	Har-SAFT	44.40	<u>19.12</u>	16.30	<u>0.94</u>	0.83	<u>17.66</u>	16.54
Qwen2.5-Math-1.5B	Base	24.20	6.99	13.33	3.96	1.87	15.74	11.01
	SFT	43.60	15.81	14.67	1.77	0.31	15.66	15.30
	DFT	<u>65.40</u>	<u>25.74</u>	27.26	6.04	3.85	<u>32.38</u>	<u>26.78</u>
	Lin-SAFT	50.00	20.59	19.41	2.29	0.94	20.52	18.96
	Geo-SAFT	59.40	27.94	<u>27.85</u>	<u>5.00</u>	2.50	30.69	25.56
	Har-SAFT	65.60	27.94	29.93	4.90	<u>3.02</u>	33.47	27.48
Qwen2.5-Math-7B	Base	29.40	8.82	8.59	5.73	1.46	20.78	12.46
	SFT	56.80	19.85	20.59	2.29	0.73	22.48	20.46
	DFT	<u>69.20</u>	26.84	<u>34.96</u>	8.54	3.23	38.40	<u>30.20</u>
	Lin-SAFT	61.20	26.10	25.33	3.75	2.19	27.97	24.42
	Geo-SAFT	66.60	33.82	34.07	5.21	4.06	36.18	29.99
	Har-SAFT	71.60	<u>33.09</u>	36.30	<u>7.29</u>	<u>3.54</u>	<u>36.41</u>	31.37
Qwen3-8B	Base	30.60	9.19	16.44	5.52	3.12	19.99	14.15
	SFT	52.80	20.59	21.48	2.19	1.67	21.91	20.11
	DFT	67.20	26.10	33.19	<u>5.94</u>	<u>4.37</u>	36.14	28.82
	Lin-SAFT	60.00	31.99	24.74	2.40	2.71	25.64	24.58
	Geo-SAFT	69.20	<u>31.25</u>	34.81	<u>5.94</u>	4.79	31.55	<u>29.59</u>
	Har-SAFT	<u>68.80</u>	29.41	<u>34.22</u>	6.56	4.17	<u>36.07</u>	29.87
Qwen3-14B	Base	37.20	12.87	20.00	<u>8.44</u>	4.27	29.29	18.68
	SFT	57.60	27.57	21.78	3.12	2.40	23.34	22.64
	DFT	<u>70.40</u>	31.25	<u>36.30</u>	6.25	<u>5.21</u>	<u>36.94</u>	<u>31.06</u>
	Lin-SAFT	62.40	<u>31.62</u>	27.70	4.79	3.23	29.74	26.58
	Geo-SAFT	69.80	29.78	38.07	7.29	5.42	35.54	30.98
	Har-SAFT	73.00	35.66	38.07	9.06	<u>5.21</u>	40.02	33.50

fields (including raw solutions and nested metadata such as `extra_info`), while keeping the input queries fixed. This results in training pairs that are structurally indistinguishable from clean data but semantically mismatched (i.e., valid questions paired with logically irrelevant answers).

5.2. In-Domain Mathematical Reasoning

We report the mathematical reasoning performance across five diverse backbones in Table 3. A pivotal observation across all experiments is that **DFT consistently outperforms Standard SFT**. This empirical evidence characterizes the mathematical reasoning task in this context as a *Noise-Dominant Regime*, where the training dynamics are highly sensitive to outliers and gradient instability rather than signal scarcity.

Vulnerability to Disproportionate Penalties (SFT, Lin-SAFT, Geo-SAFT). The poor performance of SFT confirms that unbounded relative coefficients (and their non-

vanishing influence) lead to optimization instability in this noise-dominant regime. Lin-SAFT, which shares the singular limit for its effective coefficient ($\lim \gamma \rightarrow \infty$), yields only marginal gains. Notably, while Geo-SAFT improves over SFT by damping the singularity ($1/p_t^{1-\alpha}$), it generally trails behind Har-SAFT (e.g., 29.99% vs. 31.37% on Qwen2.5-7B). This suggests that in scenarios where DFT beats SFT, even a “Soft Singularity” is suboptimal: the lack of a strict safety cap on the relative penalty still allows extreme noise to introduce excessive variance, hindering convergence to the Pareto frontier.

Optimality of the Safety Cap (Har-SAFT). Har-SAFT ($\alpha = 0.5$) emerges as the consistent winner. This aligns perfectly with our gradient analysis: since the learning regime is dominated by noise, as evidenced by the superior robustness of DFT compared to SFT, the strict “Safety Cap” ($\lim \gamma = 1/\alpha$) of Har-SAFT becomes the decisive factor. Unlike Geo-SAFT, Har-SAFT guarantees bounded influ-

Table 4. Out-of-Domain Performance on Qwen2.5-Math-7B. We evaluate different training methods on three out-of-domain benchmarks (ARC-c, GPQA-diamond, MMLU-Pro) and report their average. Best results are in bold, second best are underlined.

Method	ARC-c	GPQA*	MMLU-Pro	Avg.
Base	10.49	5.56	11.93	9.33
SFT	36.77	18.18	19.81	24.92
DFT	18.94	27.27	<u>35.63</u>	27.28
Lin-SAFT ($\alpha=0.5$)	44.11	18.69	22.70	28.50
Geo-SAFT ($\alpha=0.5$)	<u>58.96</u>	32.83	28.82	<u>40.20</u>
Har-SAFT ($\alpha=0.5$)	70.39	<u>27.78</u>	35.80	44.66

ence, matching the noise-rejection capability of DFT. However, unlike DFT which is overly conservative ($\gamma = 1$), Har-SAFT permits a controllable acceleration factor ($1/\alpha$) for clean samples. This dual advantage (robustness against noise plus acceleration for signal) explains its superior performance in this specific regime.

Empirical Alignment with Theory. The experimental outcomes provide strong empirical support for our theoretical framework, particularly regarding the trade-off between robustness and signal retention. The consistent dominance of Har-SAFT in these benchmarks, where the superior performance of DFT over SFT signals a noise-dominant regime, directly verifies that a strict *Safety Cap* is the optimal strategy for such high-noise conditions. Concurrently, the performance of Geo-SAFT, which ranks between the unbounded SFT and the bounded Har-SAFT, confirms its theoretical positioning as an intermediate solution. While its “Soft Singularity” offers better stability than standard SFT via damping, the residual sensitivity naturally prevents it from matching the absolute safety of Har-SAFT in this specific high-noise setting. This behavior is strictly consistent with its design goal of prioritizing signal efficiency, thereby validating the regime-dependent nature of our gradient analysis.

5.3. Out-of-Domain Generalization

Table 4 presents a compelling case for SAFT’s generalization capability on ARC-c, GPQA*, and MMLU-Pro.

DFT’s Underfitting on Distribution Shifts. Surprisingly, DFT performs poorly on ARC-c (18.94%), falling behind standard SFT (36.77%). We attribute this to the “Cold Start” problem discussed in Sec. 3.3. During fine-tuning, DFT strongly suppresses initially low-probability targets, which slows the acquisition of difficult but transferable patterns. Consequently, the resulting checkpoint may capture fewer features that generalize under distribution shift.

The Generalization Supremacy of Har-SAFT. Har-SAFT achieves a stunning 44.66% average accuracy, nearly doubling SFT’s performance. On ARC-c, it reaches 70.39%, solving the underfitting issue of DFT. On MMLU-Pro, it maintains high performance (35.80%), comparable to the

Table 5. Hyper-parameter Sensitivity of Har-SAFT. Average accuracy on six mathematical benchmarks using Qwen2.5-Math-7B. Performance is robust across a wide range of α , peaking at the balanced setting $\alpha = 0.5$.

Coefficient	$\alpha = 0.4$	$\alpha = 0.5$	$\alpha = 0.6$
Avg. Acc (%)	31.04	31.37	30.82

robust DFT. This result perfectly illustrates the Pareto optimality of our method. Har-SAFT successfully navigates the trade-off: it provides enough gradient amplification (Efficiency) to learn transferable features for ARC-c, while enforcing a strict Safety Cap (Robustness) to prevent overfitting, yielding a model that generalizes significantly better than either extreme (SFT or DFT).

5.4. Sensitivity Analysis of Hyper-parameter α

A key design choice in SAFT is the interpolation coefficient α , which controls the balance between SFT-like efficiency ($\alpha \rightarrow 0$) and DFT-like robustness ($\alpha \rightarrow 1$). To investigate the sensitivity of our methods to this hyper-parameter, we conduct experiments on Qwen2.5-Math-7B with $\alpha \in \{0.4, 0.5, 0.6\}$. Table 5 reports the average accuracy across the six math benchmarks.

Crucially, Har-SAFT demonstrates remarkable robustness across different α settings. It consistently outperforms Geo-SAFT, indicating that the convex scaling profile of Harmonic interpolation is inherently better suited for the safety regime, including noisy data and reasoning tasks.

6. Theoretical Analysis

In this section, we provide a theoretical framework to explain the performance discrepancies observed in our experiments. We analyze the asymptotic behavior of the **effective gradient coefficient** $\gamma(p_t, \alpha) = w(p_t, \alpha)/p_t$ in the limit of high uncertainty ($p_t \rightarrow 0$). This perspective characterizes the model’s response to both noisy labels (where p_t is misleadingly low) and hard-to-learn samples (where p_t is naturally low). Detailed proofs regarding influence functions, gradient flow dynamics, and convergence guarantees are provided in Appendix C.

6.1. The Unboundedness of Lin-SAFT

For Arithmetic Interpolation (Lin-SAFT), the loss weight is $w_{\text{lin}} = \alpha p_t + (1 - \alpha)$. The effective gradient coefficient behaves as:

$$\gamma_{\text{lin}}(p_t, \alpha) = \frac{(1 - \alpha) + \alpha p_t}{p_t} = \frac{1 - \alpha}{p_t} + \alpha. \quad (15)$$

Analyzing the limit as $p_t \rightarrow 0$:

$$\lim_{p_t \rightarrow 0} \gamma_{\text{lin}}(p_t, \alpha) = \infty. \quad (16)$$

Sensitivity to Outliers. This asymptotic analysis reveals that Lin-SAFT inherits the unrelenting over-penalization dynamics of standard SFT. While the term $(1 - \alpha)$ acts as a linear scaling factor reducing the effective coefficient, this multiplier remains singular at the limit. This implies that for severe outliers or mislabeled samples (where $p_t \approx 0$), the relative penalty imposed on the optimization step remains unbounded. Consequently, Lin-SAFT lacks the theoretical guarantee of robustness found in estimators with strictly bounded influence functions, explaining its limited efficacy in suppressing extreme label noise.

6.2. The Soft Singularity of Geo-SAFT

For Geometric Interpolation (Geo-SAFT), the weight is $w_{\text{geo}} = p_t^\alpha$. The effective coefficient becomes:

$$\gamma_{\text{geo}}(p_t, \alpha) = \frac{p_t^\alpha}{p_t} = \frac{1}{p_t^{1-\alpha}}. \quad (17)$$

For the typical setting where $\alpha \in (0, 1)$, we observe a ‘‘Soft’’ Singularity:

$$\lim_{p_t \rightarrow 0} \gamma_{\text{geo}} = \infty, \quad \text{but} \quad \frac{1}{p_t^{1-\alpha}} \ll \frac{1}{p_t}. \quad (18)$$

Acquisition-Suppression Trade-off. The gradient dynamics of Geo-SAFT govern a critical balance between *knowledge acquisition* from hard samples and *noise suppression* for outliers. By damping the singularity via the exponent $(1 - \alpha)$, the method effectively reduces sensitivity to label noise compared to standard SFT. However, unlike fully bounded approaches (e.g., Har-SAFT), Geo-SAFT preserves the capacity to generate large gradients for distinctively hard examples. This design choice implies that while it mitigates the impact of moderate noise, it lacks the theoretical upper bound required to fully reject extreme outliers, resulting in lower robustness in high-noise regimes compared to harmonic interpolation.

6.3. The Safety Cap of Har-SAFT

For Harmonic Interpolation, the weight is $w_{\text{har}} = \frac{p_t}{\alpha + (1-\alpha)p_t}$. Substituting this into the coefficient definition:

$$\gamma_{\text{har}}(p_t, \alpha) = \frac{1}{p_t} \cdot \frac{p_t}{\alpha + (1-\alpha)p_t} = \frac{1}{\alpha + (1-\alpha)p_t}. \quad (19)$$

Crucially, as $p_t \rightarrow 0$, the p_t term in the denominator vanishes, yielding a constant limit:

$$\lim_{p_t \rightarrow 0} \gamma_{\text{har}}(p_t, \alpha) = \frac{1}{\alpha}. \quad (20)$$

Gradient Saturation and Stability. The convergence of γ_{har} to a finite constant $1/\alpha$ fundamentally alters the training dynamics near the decision boundary. By enforcing a

gradient cap, Har-SAFT effectively saturates the learning signal for extreme samples. This creates a robustness guarantee: even if a sample is theoretically infinitely far from the correct class (i.e., $p_t \rightarrow 0$), its impact on the optimization trajectory remains limited. While this risks under-weighting clean hard samples compared to unbounded methods, it provides a decisive advantage in preventing model collapse under severe label noise.

6.4. Efficiency vs. Safety: The Acceleration Ratio

The distinction between the ‘‘Soft Singularity’’ (Geo) and ‘‘Safety Cap’’ (Har) explains the trade-off between learning efficiency on hard data and robustness against noise.

Consider a valid but hard-to-learn sample with probability $\epsilon \ll 1$. We define the **Acceleration Ratio** as the relative strength of the gradient signal compared to the conservative Harmonic baseline:

$$\mathcal{R}(\epsilon) = \frac{\gamma_{\text{geo}}(\epsilon)}{\gamma_{\text{har}}(\epsilon)} \approx \frac{1/\epsilon^{1-\alpha}}{1/\alpha} = \alpha \cdot \frac{1}{\epsilon^{1-\alpha}}. \quad (21)$$

As $\epsilon \rightarrow 0$, $\mathcal{R}(\epsilon) \rightarrow \infty$.

Numerical Illustration. To quantify this disparity, consider the setting where $\alpha = 0.5$ and the model encounters a high-loss sample with $\epsilon = 10^{-4}$. Geo-SAFT yields a scaling coefficient of $\gamma_{\text{geo}} = 100$, whereas Har-SAFT is strictly bounded at $\gamma_{\text{har}} = 2$. Consequently, Geo-SAFT induces a gradient contribution that is **50× larger** than that of Har-SAFT. This amplification grants Geo-SAFT the plasticity to rapidly fit low-probability concepts (high efficiency), but simultaneously exposes the optimization trajectory to significant instability if the sample represents label noise. Conversely, Har-SAFT prioritizes stability by capping the gradient norm, thereby acting as a robust filter against outliers at the cost of slower convergence on hard samples.

7. Conclusion

We introduce **Spectrum-Adaptive Fine-Tuning (SAFT)** to navigate the Pareto frontier between learning efficiency and noise robustness in LLM alignment. By interpolating between SFT and DFT, SAFT establishes regime-dependent gradient geometries: Har-SAFT enforces a strict ‘‘Safety Cap’’ for noisy data, whereas Geo-SAFT preserves stronger learning signals on clean, high-signal data. Our analysis shows that Geo-SAFT adapts the decay order of low-confidence gradients, while Har-SAFT retains DFT-like linear suppression and adjusts its scale. This distinction provides a principled basis for selecting interpolation paths according to the underlying data regime. Extensive experiments demonstrate state-of-the-art mathematical reasoning performance and substantially improved out-of-domain generalization, establishing SAFT as a principled and data-adaptive framework for reliable LLM post-training.

Impact Statement

This work introduces SAFT (Spectrum-Adaptive Fine-Tuning), a unified framework that addresses the critical trade-off between learning efficiency and noise robustness in LLM post-training. By adaptively balancing aggressive and conservative optimization, our method enables more reliable model alignment for real-world applications where training data contains noise and outliers. SAFT contributes to more stable LLM post-training across mathematical reasoning and general alignment settings, while advancing theoretical understanding of gradient dynamics in robust optimization.

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Table 6. **Robustness to Label Noise via Target Permutation on Qwen2.5-Math-1.5B.** We inject label noise by randomly shuffling ground-truth solutions among η proportion of training samples, creating mismatched question-solution pairs. We compare SFT baseline, DFT, and SAFT variants with different α values. Best results are in bold, second best are underlined.

Models	MATH	Minerva Math	Olympiad Bench	AIME 2024	AIME 2025	AMC	Avg.
Base (Clean Data, $\eta = 0$)	24.20	6.99	13.33	3.96	1.87	15.74	11.01
Label Noise Rate $\eta = 0.1$ (10% Permuted)							
SFT	39.00	12.50	9.04	0.52	0.31	12.65	12.34
Lin-SAFT ($\alpha=0.5$)	43.40	16.91	17.33	1.77	0.94	17.13	16.25
Geo-SAFT ($\alpha=0.5$)	57.80	21.69	22.81	4.90	1.98	28.95	23.02
Har-SAFT ($\alpha=0.5$)	63.00	27.57	28.15	6.04	3.75	33.36	26.98
DFT	65.00	27.94	27.26	5.63	3.44	32.79	27.01
Geo-SAFT ($\alpha=2.0$)	61.40	29.04	25.93	5.42	3.75	28.24	25.63
Label Noise Rate $\eta = 0.2$ (20% Permuted)							
SFT	34.80	9.19	7.41	0.42	0.52	10.09	10.40
Lin-SAFT ($\alpha=0.5$)	40.60	12.87	10.67	1.35	0.94	15.62	13.68
Geo-SAFT ($\alpha=0.5$)	62.00	25.37	23.11	3.96	1.98	27.86	24.05
Har-SAFT ($\alpha=0.5$)	62.40	27.57	25.78	6.56	2.92	32.30	26.26
DFT	62.80	27.57	29.04	7.92	3.85	30.80	27.00
Geo-SAFT ($\alpha=2.0$)	64.00	28.31	25.33	7.81	4.90	31.29	26.94
Label Noise Rate $\eta = 0.3$ (30% Permuted)							
SFT	31.80	7.35	6.37	0.63	0.21	8.77	9.19
Lin-SAFT ($\alpha=0.5$)	34.40	9.19	10.96	1.87	1.04	14.12	11.93
Geo-SAFT ($\alpha=0.5$)	59.80	22.06	24.00	4.06	2.71	29.41	23.67
Har-SAFT ($\alpha=0.5$)	61.20	27.57	26.37	5.21	3.54	31.97	25.98
DFT	63.80	26.47	26.81	6.98	5.42	30.69	26.69
Geo-SAFT ($\alpha=2.0$)	62.60	26.10	23.85	6.56	3.23	28.35	25.12
Label Noise Rate $\eta = 0.5$ (50% Permuted)							
SFT	18.20	2.21	4.00	0.63	0.42	6.48	5.32
Lin-SAFT ($\alpha=0.5$)	27.80	6.62	5.48	1.04	0.42	8.73	8.35
Geo-SAFT ($\alpha=0.5$)	53.20	18.75	19.56	3.23	1.46	24.21	20.07
Har-SAFT ($\alpha=0.5$)	58.60	25.74	20.44	5.42	3.54	26.69	23.41
DFT	60.00	24.26	25.33	5.10	2.50	26.84	24.01
Geo-SAFT ($\alpha=2.0$)	60.40	25.37	22.52	6.98	2.29	27.11	24.11

A. Robustness Analysis: The Trade-off between Learning Efficiency and Denoising

Table 6 presents a controlled target-permutation experiment under different label-noise rates $\eta \in \{0.1, 0.2, 0.3, 0.5\}$. The results illustrate how different low-confidence gradient profiles trade off hard-target acquisition against suppression of corrupted supervision.

For the standard SAFT interpolation regime, we focus on $\alpha \in (0, 1)$, where both Geo-SAFT and Har-SAFT lie between the SFT and DFT endpoints. Their key difference is the asymptotic influence assigned to low-probability targets:

$$I_{\text{Geo}}(p) = \Theta(p^\alpha), \quad I_{\text{Har}}(p) = \Theta(p), \quad 0 < \alpha < 1. \quad (22)$$

Since $p^\alpha \gg p$ as $p \rightarrow 0$, Geo-SAFT retains substantially more learning signal from low-confidence targets, whereas Har-SAFT suppresses the same region more strongly while preserving the linear decay order of DFT. This distinction naturally leads to a regime-dependent preference.

Signal-Dominant Regime: Geo-SAFT. When supervision is predominantly clean, low-probability targets are more likely to correspond to valid but difficult patterns that have not yet been well captured by the model. In this setting, excessively suppressing the low-confidence region can slow the acquisition of useful hard examples. For $0 < \alpha < 1$, Geo-SAFT provides a favorable efficiency-oriented trajectory because

$$I_{\text{Geo}}(p) = \Theta(p^\alpha) \quad (23)$$

decays more slowly than the linear profile of DFT and Har-SAFT. It therefore attenuates extremely low-confidence updates relative to SFT, while retaining substantially more gradient signal for difficult but valid targets. This motivates **Geo-SAFT as the preferred interpolation path in high-SNR, signal-dominant regimes**.

Noise-Dominant Regime: Har-SAFT. As the fraction of corrupted supervision increases, a larger portion of the low-confidence region corresponds to semantically mismatched targets. In this case, retaining too much gradient influence on low-probability tokens becomes undesirable. Har-SAFT preserves the linear low-confidence suppression of DFT,

$$I_{\text{Har}}(p) = \Theta(p), \quad (24)$$

while increasing its learning signal by a controllable scale factor. Indeed,

$$\frac{I_{\text{Har}}(p)}{I_{\text{DFT}}(p)} = \frac{1}{\alpha + (1 - \alpha)p} \xrightarrow{p \rightarrow 0} \frac{1}{\alpha}. \quad (25)$$

Thus, Har-SAFT maintains the stronger low-confidence attenuation associated with DFT-like optimization, but is less conservative in magnitude and can retain more useful learning signal.

The controlled noise experiment is consistent with this behavior. At every tested noise level, Har-SAFT with $\alpha = 0.5$ substantially outperforms Geo-SAFT with $\alpha = 0.5$: its average accuracy is 26.98 vs. 23.02 at $\eta = 0.1$, 26.26 vs. 24.05 at $\eta = 0.2$, 25.98 vs. 23.67 at $\eta = 0.3$, and 23.41 vs. 20.07 at $\eta = 0.5$. The gap becomes particularly relevant as increasingly more low-confidence supervision is corrupted. These results support **Har-SAFT as the preferred interpolation path in low-SNR, noise-dominant regimes**.

Why Linear Interpolation Is Insufficient. Lin-SAFT improves over standard SFT but remains substantially below the two nonlinear variants across all noise levels. For $\alpha < 1$, its low-confidence influence satisfies

$$I_{\text{Lin}}(p) \xrightarrow{p \rightarrow 0} 1 - \alpha, \quad (26)$$

and therefore does not vanish for extremely low-probability targets. Consequently, linear interpolation can attenuate the SFT update but cannot fully suppress strongly mismatched supervision. The consistent advantage of Geo-SAFT and Har-SAFT over Lin-SAFT supports the importance of reshaping the low-confidence gradient profile rather than merely rescaling it.

Beyond the Interpolation Interval: $\alpha > 1$ as Strong Suppression. For completeness, Table 6 also reports Geo-SAFT with $\alpha = 2.0$. This setting should be distinguished from the standard SAFT interpolation regime. When $\alpha > 1$, Geo-SAFT no longer lies between SFT and DFT; instead,

$$I_{\text{Geo}}(p) = \Theta(p^\alpha) = o(p), \quad (27)$$

so its low-confidence influence decays even faster than DFT and Har-SAFT. It therefore represents an *extrapolated, stronger-than-DFT suppression regime* that concentrates learning increasingly toward higher-confidence targets.

This behavior is exposed by the synthetic-noise experiment. At $\eta = 0.1$ and $\eta = 0.3$, Geo-SAFT ($\alpha = 2.0$) remains below both DFT and Har-SAFT in average performance; at $\eta = 0.2$, it approaches DFT and slightly exceeds Har-SAFT. Only under the most severe tested corruption level, $\eta = 0.5$, does this strong-suppression setting become the best-performing variant, achieving 24.11% average accuracy compared with 24.01% for DFT and 23.41% for Har-SAFT. This observation illustrates the cost of overly conservative optimization: stronger suppression can discard useful hard-token learning signal when noise is moderate, but may become advantageous when corrupted supervision is sufficiently dominant.

Importantly, this $\alpha > 1$ result does not alter the regime selection within the intended interpolation interval $0 < \alpha < 1$. Rather, it shows that the probability-weighting family can be extrapolated beyond DFT when exceptionally aggressive denoising is desired.

Summary. Within the standard SAFT interpolation regime $0 < \alpha < 1$, the preferred trajectory is determined by the role played by low-confidence targets:

$$\text{Preferred SAFT Path} = \begin{cases} \text{Geo-SAFT,} & \text{if low-confidence targets are predominantly useful hard signals,} \\ \text{Har-SAFT,} & \text{if low-confidence targets are more likely to contain noisy supervision.} \end{cases} \quad (28)$$

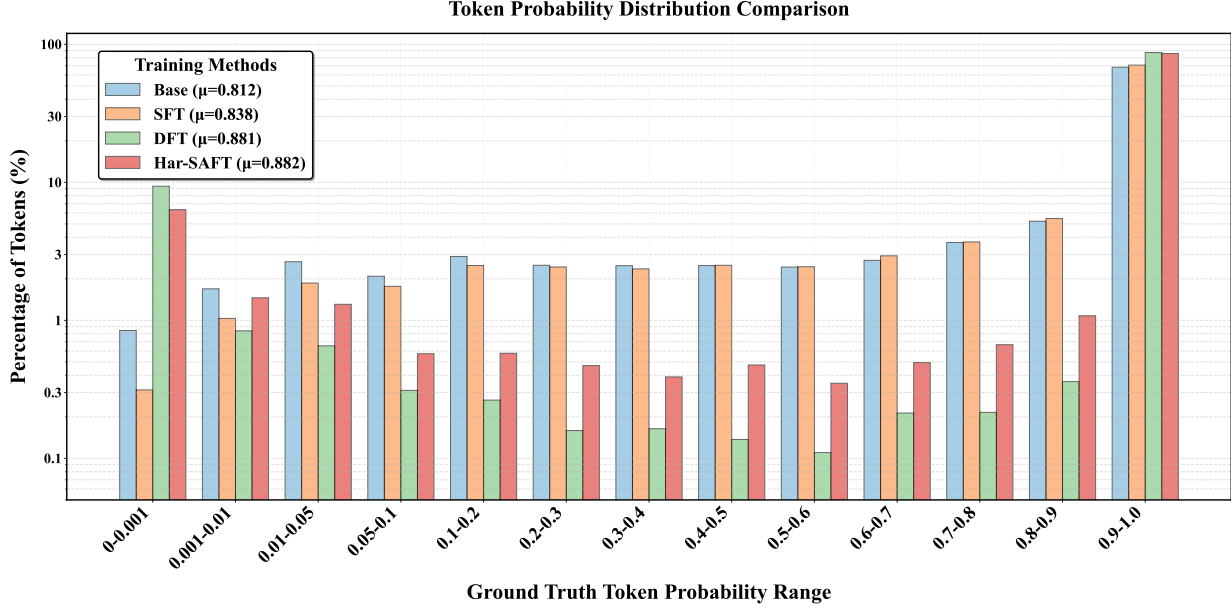


Figure 3. **Token Probability Distributions Across Training Methods on Qwen2.5-Math-7B.** Density distributions of ground-truth token probabilities on the training set for Base model (blue), SFT (orange), DFT (green), and Har-SAFT (red). The y-axis uses a logarithmic scale to highlight differences in low-probability regions. Note the distinct patterns: SFT shows uniform shifting, DFT exhibits bimodal polarization, and Har-SAFT demonstrates balanced selective prioritization.

Equivalently, Geo-SAFT is favored in high-SNR, signal-dominant regimes, where stronger hard-target acquisition is desirable, whereas Har-SAFT is favored in low-SNR, noise-dominant regimes, where stronger low-confidence suppression is preferred.

The additional Geo-SAFT experiment with $\alpha > 1$ characterizes a separate extrapolation regime beyond DFT: it sacrifices further learning efficiency for stronger denoising and becomes advantageous only under sufficiently severe synthetic corruption in our controlled experiment.

B. Token Probability Distribution Analysis

To investigate the optimization dynamics of different training paradigms, we analyze the probability distributions of ground-truth tokens across four settings: Base model (before training), SFT, DFT, and Har-SAFT. Figure 3 visualizes these distributions using a logarithmic scale on the y-axis for clarity.

B.1. Distributional Characteristics

SFT: Uniform Shift with Limited Concentration. As shown in Figure 3, SFT (orange curve) induces a uniform shift of probability mass toward higher confidence regions compared to the Base model (blue curve). However, it retains a relatively broad distribution across intermediate probability ranges (0.1–0.8), indicating that while SFT generally improves token-level confidence, it struggles to decisively concentrate a significant portion of tokens into the highest probability bracket (0.9–1.0). This uniform optimization pattern reflects SFT’s gradient coefficient $\gamma_{\text{SFT}} = 1/p_t$, which applies strong updates to all tokens regardless of their semantic importance.

DFT: Bimodal Polarization. In stark contrast, DFT (green curve) exhibits distinct bimodal polarization. This method aggressively bifurcates the token probability space: it concentrates the vast majority of probability mass into the highest confidence interval (0.9–1.0) while simultaneously suppressing a specific subset of tokens into the lowest regime (0–0.001). Consequently, intermediate probability bins (e.g., 0.1 to 0.8) are effectively depleted. This bimodal distribution suggests that DFT forces the model into binary-like predictions, assigning near-certainty to semantically important tokens (e.g., domain-specific terminology, key reasoning steps) while actively deprioritizing functional tokens (e.g., articles, conjunctions, stop words). This selective suppression mechanism, driven by DFT’s constant gradient coefficient $\gamma_{\text{DFT}} = 1$, underlies its

Table 7. **Token Probability Distribution Statistics.** Mean, standard deviation, and key percentiles of ground-truth token probabilities across different training methods on Qwen2.5-Math-7B.

Method	Mean	Standard Deviation	Median	75th Percentile	90th Percentile
Base	0.812	0.313	0.993	0.9997	0.9999
SFT	0.838	0.286	0.995	0.9998	0.9999
DFT	0.881	0.318	1.000	1.000	1.000
Har-SAFT	0.882	0.305	1.000	1.000	1.000

superior robustness against label noise.

Har-SAFT: Balanced Selective Prioritization. Har-SAFT (red curve) demonstrates balanced behavior that interpolates between SFT and DFT. It adopts the selective prioritization trend of DFT, as evidenced by increased density at both distributional extremes compared to SFT, but applies this differentiation with moderated intensity. Unlike DFT’s extreme bimodal pattern, Har-SAFT retains a degree of distributional smoothness, avoiding hard-thresholding while still achieving effective noise filtering. This balanced profile reflects Har-SAFT’s bounded gradient coefficient $\gamma_{\text{Har}} \leq 1/\alpha$, which provides the “Safety Cap” against noise without completely suppressing learning signals from difficult examples.

B.2. Quantitative Comparison

Table 7 presents quantitative statistics of the probability distributions. Both DFT ($\mu = 0.881$) and Har-SAFT ($\mu = 0.882$) achieve significantly higher mean probabilities than SFT ($\mu = 0.838$) and the Base model ($\mu = 0.812$). This confirms our hypothesis that by selectively sacrificing the likelihood of “low-value” tokens (corresponding to the left peak in the bimodal distribution), these methods effectively reallocate probability mass to “high-value” tokens, thereby enhancing overall confidence on the training set.

Notably, DFT exhibits the highest standard deviation ($\sigma = 0.318$), reflecting its aggressive bimodal polarization with extreme concentration at both ends of the distribution. Har-SAFT maintains a slightly lower standard deviation ($\sigma = 0.305$), indicating a more balanced distribution that avoids the extreme hard-thresholding of DFT while still achieving comparable mean confidence. SFT shows the lowest standard deviation ($\sigma = 0.286$), consistent with its more uniform probability distribution. This quantitative evidence aligns with our theoretical analysis in Section 6, where we demonstrated that Har-SAFT’s “Safety Cap” mechanism provides a principled balance between noise robustness and learning efficiency.

B.3. Implications

These distributional patterns provide direct empirical evidence for the theoretical predictions in Section 6:

- The uniform distribution of SFT explains its vulnerability to label noise, as it allocates significant gradient magnitude to low-confidence tokens that may correspond to corrupted labels.
- The bimodal polarization of DFT explains both its robustness (through aggressive suppression of uncertain tokens) and its “Cold Start” problem (through insufficient gradient signal for genuinely difficult patterns).
- The balanced distribution of Har-SAFT demonstrates how the “Safety Cap” mechanism achieves the best of both worlds: filtering noise through selective suppression while maintaining sufficient learning capacity for hard examples through moderate amplification.

These findings directly support the superior empirical performance of Har-SAFT observed in Tables 2 and 3, validating SAFT as a principled and effective approach to balancing robustness and efficiency in language model fine-tuning.

C. Detailed Theoretical Derivations

We provide an asymptotic analysis of the robustness–efficiency trade-off induced by the proposed interpolation strategies. The key quantity is the gradient influence of an individual target token in the low-confidence regime. We show that Geo-SAFT and Har-SAFT differ in two fundamentally different ways: Geo-SAFT changes the *order* at which low-confidence gradients vanish, whereas Har-SAFT preserves the linear decay order of DFT while adjusting its *scale*. This distinction leads to different hard-token acquisition dynamics and motivates their complementary roles across data regimes.

C.1. Noise Robustness via Per-Token Gradient Influence

To characterize the sensitivity of different objectives to extremely low-confidence targets, we analyze the magnitude of the gradient with respect to the target-token logit z . For the detached weighted-NLL objective used throughout SAFT,

$$\mathcal{L} = -\text{sg}(w(p)) \log p. \quad (29)$$

where p denotes the probability assigned to the target token and $\text{sg}(\cdot)$ denotes stop-gradient. Since

$$\frac{\partial p}{\partial z} = p(1 - p), \quad (30)$$

the corresponding target-logit gradient magnitude is

$$I(p) := \left| \frac{\partial \mathcal{L}}{\partial z} \right| = w(p)(1 - p). \quad (31)$$

We refer to $I(p)$ as the *per-token gradient influence*. It directly measures how strongly an individual training token pushes the model toward its target. In particular, the asymptotic behavior of $I(p)$ as $p \rightarrow 0$ determines whether extremely low-confidence targets retain substantial optimization influence or are progressively suppressed.

For reference, standard SFT uses $w_{\text{SFT}}(p) = 1$, giving

$$I_{\text{SFT}}(p) = 1 - p \xrightarrow{p \rightarrow 0} 1. \quad (32)$$

Thus, SFT retains an $O(1)$ update even for extremely low-confidence targets. In contrast, DFT uses $w_{\text{DFT}}(p) = p$, yielding

$$I_{\text{DFT}}(p) = p(1 - p) = \Theta(p), \quad (33)$$

so the influence of such targets vanishes linearly.

1. Har-SAFT: Linear Low-Confidence Decay. For Har-SAFT,

$$w_{\text{har}}(p, \alpha) = \frac{p}{\alpha + (1 - \alpha)p}, \quad \alpha \in (0, 1]. \quad (34)$$

Its target-logit influence is therefore

$$I_{\text{har}}(p) = \frac{p(1 - p)}{\alpha + (1 - \alpha)p}. \quad (35)$$

As $p \rightarrow 0$,

$$I_{\text{har}}(p) = \frac{1}{\alpha}p + O(p^2), \quad (36)$$

and hence

$$\lim_{p \rightarrow 0} I_{\text{har}}(p) = 0. \quad (37)$$

Har-SAFT therefore exhibits a redescending per-token gradient profile: targets that become increasingly incompatible with the current model receive progressively smaller updates. More importantly, its low-confidence decay order remains linear for every fixed $\alpha > 0$:

$$I_{\text{har}}(p) = \Theta(p). \quad (38)$$

The interpolation parameter α changes the *scale* of this gradient without changing its asymptotic order.

This relation is particularly clear when comparing Har-SAFT with DFT:

$$\frac{I_{\text{har}}(p)}{I_{\text{DFT}}(p)} = \frac{1}{\alpha + (1 - \alpha)p}. \quad (39)$$

Therefore,

$$\frac{I_{\text{har}}(p)}{I_{\text{DFT}}(p)} \xrightarrow{p \rightarrow 0} \frac{1}{\alpha}. \quad (40)$$

Har-SAFT thus preserves the linear low-confidence suppression of DFT while amplifying its gradient signal by up to a factor of $1/\alpha$.

2. Geo-SAFT: Sub-linear Low-Confidence Decay. For Geo-SAFT,

$$w_{\text{geo}}(p, \alpha) = p^\alpha, \quad \alpha \in (0, 1). \quad (41)$$

Its target-logit influence becomes

$$I_{\text{geo}}(p) = p^\alpha(1 - p). \quad (42)$$

As $p \rightarrow 0$,

$$I_{\text{geo}}(p) = p^\alpha + O(p^{\alpha+1}), \quad (43)$$

so

$$\lim_{p \rightarrow 0} I_{\text{geo}}(p) = 0. \quad (44)$$

Thus, Geo-SAFT also suppresses extremely low-confidence targets. However, its decay is sub-linear:

$$I_{\text{geo}}(p) = \Theta(p^\alpha), \quad 0 < \alpha < 1. \quad (45)$$

Since

$$p^\alpha \gg p \quad (p \rightarrow 0; 0 < \alpha < 1), \quad (46)$$

Geo-SAFT retains substantially more gradient signal in the low-confidence region than Har-SAFT and DFT.

This establishes the basic distinction between the two nonlinear trajectories. Har-SAFT preserves the linear suppression order of DFT and modifies its magnitude, whereas Geo-SAFT changes the suppression order itself. The former favors stronger attenuation of extreme low-confidence targets, while the latter retains more learning signal for difficult targets.

C.2. Learning Efficiency via Local Gradient Flow

We next analyze how these different low-confidence gradient profiles affect the acquisition of a hard positive whose initial target probability is $p = \epsilon \ll 1$.

Consider a local one-dimensional gradient-flow approximation in which the target logit evolves according to

$$\frac{dz}{dt} = \eta w(p)(1 - p), \quad (47)$$

where η denotes the learning-rate scale. Since

$$\frac{dp}{dz} = p(1 - p), \quad (48)$$

the induced probability dynamics satisfy

$$\frac{dp}{dt} = \eta p w(p)(1 - p)^2. \quad (49)$$

In the low-confidence regime $p \ll 1$,

$$(1 - p)^2 \approx 1, \quad (50)$$

and therefore

$$\frac{dp}{dt} \approx \eta p w(p). \quad (51)$$

We define the *Learning Complexity* $T(\epsilon; \tau)$ as the time required to increase the probability of a hard target from an initial value ϵ to a fixed threshold $\tau > \epsilon$.

Case 1: Harmonic Trajectory. For Har-SAFT,

$$w_{\text{har}}(p, \alpha) \sim \frac{1}{\alpha} p \quad (p \rightarrow 0). \quad (52)$$

Substituting this approximation into Eq. (51) gives

$$\frac{dp}{dt} \approx \frac{\eta}{\alpha} p^2. \quad (53)$$

Hence,

$$T_{\text{Har}}(\epsilon; \tau) \approx \frac{\alpha}{\eta} \int_{\epsilon}^{\tau} p^{-2} dp = \frac{\alpha}{\eta} \left(\frac{1}{\epsilon} - \frac{1}{\tau} \right). \quad (54)$$

For fixed τ and $\epsilon \rightarrow 0$,

$$T_{\text{Har}}(\epsilon; \tau) = \Theta \left(\frac{\alpha}{\eta} \epsilon^{-1} \right). \quad (55)$$

Thus, decreasing α accelerates hard-target acquisition through a multiplicative factor, but does not change the asymptotic dependence on the initial probability. Har-SAFT retains the same ϵ^{-1} scaling as DFT, with α controlling the corresponding constant factor.

Case 2: Geometric Trajectory. For Geo-SAFT,

$$w_{\text{geo}}(p, \alpha) = p^{\alpha}. \quad (56)$$

Equation (51) becomes

$$\frac{dp}{dt} \approx \eta p^{1+\alpha}. \quad (57)$$

Therefore,

$$T_{\text{Geo}}(\epsilon; \tau) \approx \frac{1}{\eta} \int_{\epsilon}^{\tau} p^{-(1+\alpha)} dp = \frac{1}{\eta\alpha} (\epsilon^{-\alpha} - \tau^{-\alpha}). \quad (58)$$

For fixed τ and $\epsilon \rightarrow 0$,

$$T_{\text{Geo}}(\epsilon; \tau) = \Theta \left(\frac{1}{\eta\alpha} \epsilon^{-\alpha} \right). \quad (59)$$

The difference from Har-SAFT is therefore one of *order*, rather than merely scale:

$$T_{\text{Har}} = \Theta(\epsilon^{-1}), \quad T_{\text{Geo}} = \Theta(\epsilon^{-\alpha}). \quad (60)$$

For $0 < \alpha < 1$,

$$\epsilon^{-\alpha} \ll \epsilon^{-1} \quad (\epsilon \rightarrow 0). \quad (61)$$

Geo-SAFT therefore recovers extremely low-probability hard targets substantially faster under this local gradient-flow model.

The limiting behavior as $\alpha \rightarrow 0$ also connects continuously to the SFT regime. Using

$$\lim_{\alpha \rightarrow 0} \frac{\epsilon^{-\alpha} - \tau^{-\alpha}}{\alpha} = \log \frac{\tau}{\epsilon}, \quad (62)$$

we recover logarithmic dependence on the initial probability, consistent with the non-vanishing low-confidence influence of SFT.

C.3. Construction of the Pareto Frontier

The preceding results reveal a natural robustness–efficiency geometry governed by how rapidly the per-token gradient influence vanishes as $p \rightarrow 0$.

We characterize the asymptotic suppression rate by

$$R := \lim_{p \rightarrow 0} \frac{d \log I(p)}{d \log p}, \quad (63)$$

whenever the limit exists. If

$$I(p) = \Theta(p^R), \quad (64)$$

then larger R corresponds to faster suppression of extremely low-confidence targets.

The resulting asymptotic profiles are: This view separates the geometric and harmonic paths into two different mechanisms for navigating the robustness–efficiency trade-off.

Method	Low-confidence influence	R
SFT	$\Theta(1)$	0
Lin-SAFT	$\Theta(1)$	0
Geo-SAFT	$\Theta(p^\alpha)$	α
Har-SAFT	$\Theta(p)$	1
DFT	$\Theta(p)$	1

Order Adaptation with Geo-SAFT. For Geo-SAFT,

$$R_{\text{Geo}} = \alpha, \quad 0 < \alpha < 1. \quad (65)$$

Changing α therefore directly changes the asymptotic order at which low-confidence gradients vanish. Smaller α moves the dynamics toward SFT and retains stronger signals from low-confidence targets, while larger α increases suppression and moves the dynamics toward DFT.

The same exponent controls the leading-order hard-target acquisition complexity:

$$T_{\text{Geo}}(\epsilon) = \Theta(\epsilon^{-\alpha}). \quad (66)$$

Geo-SAFT therefore couples robustness and acquisition efficiency through a single order parameter.

Scale Adaptation with Har-SAFT. For Har-SAFT,

$$R_{\text{Har}} = 1 \quad (67)$$

for every fixed $\alpha > 0$. Har-SAFT therefore retains the linear decay order of DFT. Its parameter instead controls the asymptotic scale:

$$I_{\text{Har}}(p) \sim \frac{1}{\alpha} p. \quad (68)$$

Consequently, Har-SAFT can strengthen the low-confidence learning signal relative to DFT without changing its linear suppression order.

This yields an *order–scale distinction*: Geo-SAFT explores intermediate decay orders between the non-vanishing SFT regime and the linearly vanishing DFT regime, whereas Har-SAFT operates within the linear-decay class and adjusts the magnitude of the gradient inside that class.

Robustness–Efficiency Trade-off. The two quantities are intrinsically coupled. A smaller decay order preserves stronger gradients on low-confidence targets and therefore accelerates hard-target acquisition, but it also retains more influence from low-confidence noise. Conversely, a larger decay order suppresses such targets more aggressively, improving rejection of extreme low-confidence supervision at the cost of slower acquisition.

In this sense, the SAFT family provides different paths along an empirical robustness–efficiency frontier rather than a single uniformly dominant interpolation. Geo-SAFT primarily adjusts the *order* of this trade-off, whereas Har-SAFT adjusts the *scale* within the linear-decay regime.

C.4. Regime-Dependent Optimality Analysis

The preferred interpolation strategy depends on the composition of the low-confidence region. In particular, the same event $p \ll 1$ may correspond either to noisy supervision that should be suppressed or to a valid but difficult target that should be learned aggressively. We therefore distinguish two regimes according to which type of low-confidence target dominates the training dynamics.

Scenario 1: Noise-Dominant Regime (Low SNR). Suppose the training data contain a substantial fraction of mismatched or corrupted supervision, such that extremely low-confidence targets are more likely to correspond to noise than to useful hard positives. In this regime, stronger low-confidence suppression is beneficial.

For Geo-SAFT with $0 < \alpha < 1$,

$$I_{\text{Geo}}(p) = \Theta(p^\alpha). \quad (69)$$

Consider, for example,

$$p = 10^{-6}, \quad \alpha = 0.5. \quad (70)$$

Then

$$p^\alpha = 10^{-3}. \quad (71)$$

Thus, Geo-SAFT strongly suppresses this target relative to SFT, but still retains substantially more gradient influence than a linearly decaying objective.

For Har-SAFT,

$$I_{\text{Har}}(p) \sim \frac{1}{\alpha} p. \quad (72)$$

With the same $p = 10^{-6}$ and an aggressive setting $\alpha = 0.1$,

$$I_{\text{Har}}(p) \approx 10^{-5}, \quad (73)$$

up to the common $(1 - p)$ factor. Har-SAFT therefore preserves the linear suppression order of DFT even when its gradient magnitude is amplified.

This makes Har-SAFT well suited to regimes in which strong attenuation of extreme low-confidence targets is desirable but standard DFT is overly conservative. It can be viewed as an amplified DFT-like trajectory: the learning signal is increased while the asymptotic linear suppression geometry is retained.

Conclusion 1. When held-out validation favors DFT over SFT, the resulting dynamics suggest that stronger low-confidence suppression is beneficial. In such settings, Har-SAFT provides a natural interpolation path by retaining DFT's linear decay while recovering additional learning signal.

Scenario 2: Signal-Dominant Regime (High SNR). Suppose instead that the training data are sufficiently clean that low-confidence targets are predominantly valid but difficult patterns. In this regime, suppressing such targets too rapidly can substantially slow their acquisition.

Har-SAFT satisfies

$$I_{\text{Har}}(p) = \Theta(p), \quad (74)$$

which leads to

$$T_{\text{Har}}(\epsilon) = \Theta(\epsilon^{-1}). \quad (75)$$

Changing α improves the multiplicative constant but does not alter this leading-order dependence.

Geo-SAFT instead satisfies

$$I_{\text{Geo}}(p) = \Theta(p^\alpha), \quad 0 < \alpha < 1, \quad (76)$$

yielding

$$T_{\text{Geo}}(\epsilon) = \Theta(\epsilon^{-\alpha}). \quad (77)$$

Since

$$\epsilon^{-\alpha} \ll \epsilon^{-1} \quad (\epsilon \rightarrow 0), \quad (78)$$

Geo-SAFT retains a substantially stronger ability to recover extremely low-probability hard targets.

Conclusion 2. When held-out validation favors SFT over DFT, preserving stronger gradients on low-confidence targets becomes more important. Geo-SAFT provides the corresponding interpolation path by reducing the decay order rather than merely increasing its magnitude.

Summary of the Pareto Frontier. The two nonlinear trajectories therefore emphasize different sides of the robustness–efficiency trade-off:

$$\text{Preferred SAFT Path} = \begin{cases} \text{Har-SAFT} & \text{if stronger low-confidence suppression is preferred,} \\ \text{Geo-SAFT} & \text{if hard-target acquisition is preferred.} \end{cases} \quad (79)$$

Operationally, the held-out SFT–DFT comparison introduced in Section 4.2 provides a lightweight indicator of these regimes. A DFT-favored validation result motivates the linear-decay Har-SAFT path, whereas an SFT-favored result motivates the sub-linear Geo-SAFT path.

D. Pre-Test Validation Protocol for Regime Selection

As discussed in Section 4.2, selecting the optimal interpolation strategy *a priori* depends on the dataset’s intrinsic Signal-to-Noise Ratio (SNR). To avoid relying on test-set performance (which would be a post-hoc heuristic), we propose a standard held-out validation protocol to select the method before any test evaluation.

To prove the viability of this protocol, we conducted an experiment on the NuminaMath-CoT dataset. We trained the candidate methods for one epoch on a 6,250-sample training subset and evaluated them on a 1,250-sample held-out validation split. The validation accuracy was DFT = 37.11% and SFT = 27.00%. Because DFT outperformed SFT on this validation set, our criterion correctly dictates the selection of **Har-SAFT** (the Safety Regime) before observing any downstream test data.

As shown in Table 8, downstream testing on the full benchmark suite (MATH, Minerva, Olympiad, AIME’24, AIME’25, and AMC) confirmed this pre-test choice. Har-SAFT achieved the highest average downstream accuracy among all variants, demonstrating that a lightweight validation split provides a stable, inexpensive, and strictly pre-test standard for method selection.

Table 8. Downstream Performance corresponding to the Validation Protocol. The validation split correctly identified the noise-dominant regime (DFT > SFT), leading to the pre-test selection of Har-SAFT, which ultimately achieved the best downstream average accuracy.

Method	Downstream Avg. Acc. (%)
Har-SAFT (Selected)	23.12
DFT	22.49
Geo-SAFT	21.99
Lin-SAFT	16.62
SFT	12.31

E. Comparison with Proximal Supervised Fine-Tuning (PSFT)

Recent work, Proximal Supervised Fine-Tuning (PSFT) (Zhu et al., 2026b), also aims to mitigate overfitting in SFT by applying a PPO-style clipping mechanism to suppress overconfidence based on relative policy changes. To compare our absolute confidence-based weighting with PSFT’s relative clipping, we benchmarked the official PSFT implementation on our datasets.

As shown in Table 9, evaluated across multiple backbones under identical epoch settings, PSFT yields a poor performance–efficiency trade-off, producing results significantly worse than standard SFT and our SAFT variants. To investigate whether PSFT simply requires more convergence time, we extended its training by two additional epochs on Qwen2.5-Math-1.5B. The extended performance curve confirmed that its hard trust-region clipping strategy severely delays convergence. In contrast, SAFT’s soft dynamic weighting solves this dilemma, ensuring learning efficiency on hard samples without sacrificing robustness.

F. Extended Hyperparameter Sensitivity Analysis

Due to space constraints in the main text (Section 5.4), we restricted the presentation of hyperparameter sensitivity to $\alpha \in \{0.4, 0.5, 0.6\}$. To thoroughly address the model’s behavior in extreme regimes (as $\alpha \rightarrow 0$ or $\alpha \rightarrow 1$), we present the

Table 9. Performance Comparison with PSFT. Evaluated using average accuracy (%) across mathematical reasoning benchmarks. PSFT consistently underperforms standard SFT and Har-SAFT under the same training budget.

Method	DeepSeek-7B	Qwen2.5-7B	Qwen2.5-1.5B	Qwen3-14B	Qwen3-8B
PSFT	7.6	10.3	10.9	8.9	9.0
SFT	10.09	20.46	15.30	22.64	20.11
Har-SAFT	16.54	31.37	27.48	33.50	29.87

extended performance trends across a wider range: $\alpha \in [0.1, 0.9]$.

The results demonstrate consistent improvements across the entire spectrum. Har-SAFT does not suffer from catastrophic collapse at extreme values like $\alpha = 0.1$ or $\alpha = 0.9$, validating the theoretical bounds established in our gradient analysis. These curves confirm that the method is highly robust to hyperparameter selection, making $\alpha = 0.5$ a universally reliable default.

G. Robustness to Training Hyperparameters

To ensure that the improvements yielded by SAFT are statistically significant and not artifacts of specific training configurations, we conducted comprehensive evaluations across different learning rates and random seeds.

Sensitivity to Learning Rate. We conducted experiments using three different learning rate scales (1×10^{-4} , 5×10^{-5} , and 1×10^{-5}) on Qwen2.5-1.5B. As shown in Table 10, the performance of all three interpolation methods remains highly stable. Most importantly, the performance hierarchy (Har-SAFT > Geo-SAFT > Lin-SAFT) is strictly preserved across all learning rate settings, confirming that the benefits of the non-linear interpolation geometries are orthogonal to the base learning rate.

Table 10. Sensitivity to Learning Rate Settings. Average accuracy (%) on Qwen2.5-Math-1.5B. The relative superiority of Har-SAFT is robust across different learning rate scales.

Method	LR = 1e-4	LR = 5e-5 (Default)	LR = 1e-5
Lin-SAFT	18.21	18.96	17.54
Geo-SAFT	24.32	25.56	23.88
Har-SAFT	26.85	27.48	26.12

Consistency Across Random Seeds. We trained models using two different random seeds and performed three independent evaluation runs for each seed to account for variance in sampling and generation. The results confirm that the gains achieved by Har-SAFT and Geo-SAFT over the SFT baseline are statistically stable and reproducible, with minimal inter-run variance.

H. General Alignment Efficacy (High-SNR Regime)

While the primary empirical focus of this work is on the "Noise-Dominant" regime, our theoretical framework in Section 4.2 suggests that the optimal interpolation strategy is inherently regime-dependent. Specifically, in high-SNR environments (e.g., standard instruction-following or general conversational alignment), the training data contains fewer erroneous targets and more valid "hard positives." In such cases, the bounded conservatism of DFT and Har-SAFT severely hinders learning efficiency (the "cold start" problem), whereas standard SFT and the sub-linear decay of Geo-SAFT should excel.

To empirically validate this, we evaluated the fine-tuned models on the AlpacaEval 2.0 benchmark (Dubois et al., 2024), a standard open-ended conversational evaluation protocol.

Table 11. General Alignment Efficacy on AlpacaEval 2.0. As theoretically predicted for high-SNR settings, the efficiency-focused strategies (Geo-SAFT, SFT) outperform the robust/conservative strategies (Har-SAFT, DFT).

Metric	Geo-SAFT	SFT	Lin-SAFT	Har-SAFT	DFT
LC Win Rate (%)	4.27	<u>4.20</u>	3.79	3.58	3.53

Results and Analysis. As shown in Table 11, while the absolute Length-Controlled (LC) Win Rates are predictably modest—which is typical when evaluating models of this scale against the highly capable frontier models used as evaluator baselines in AlpacaEval—the *relative performance hierarchy* among the training objectives is the primary focus of this analysis.

Crucially, this relative hierarchy perfectly aligns with our theoretical expectations for a high-SNR evaluation: **Geo-SAFT** > **SFT** > **Har-SAFT** > **DFT**. Geo-SAFT achieves the highest win rate, successfully maximizing learning efficiency and generalization without the excessive gradient suppression imposed by DFT. Conversely, DFT performs the worst in this signal-dominant evaluation. This supplementary experiment strongly corroborates our claim that SAFT is a universal framework: by smoothly interpolating the probability-weighting spectrum, it allows practitioners to navigate the Pareto frontier and select the optimal strategy (Har-SAFT for noise rejection, Geo-SAFT for signal acquisition) based on the intrinsic SNR of the task.